Techinical details on the implementation

September 11, 2026

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1 Theory

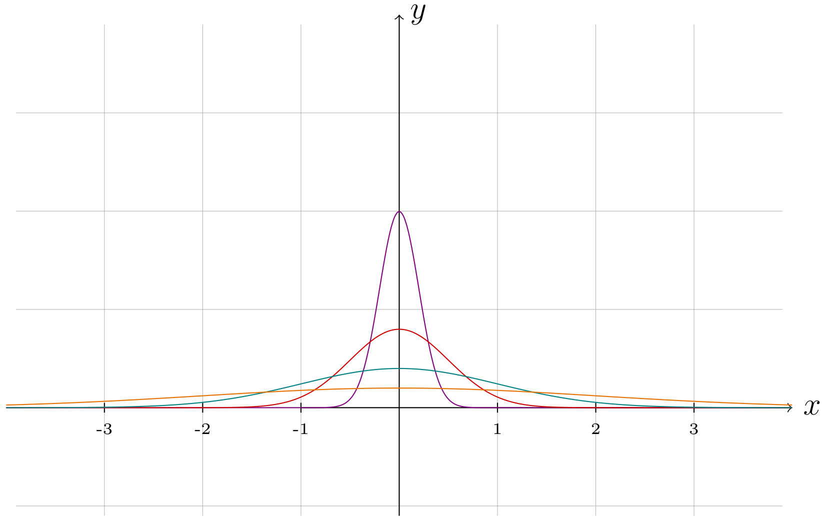

Figure 1: Normal PDF $\frac{1}{\sqrt{2\pi}\sigma}e^{-\frac{(x-\mu)^2}{2\sigma^2}}$

1.1 Central limit theorem

For $X_1, ..., X_n \overset{iid}{\sim} X$ we have:

$$\frac{\sum_{i=1}^{n} X_i - n\,\mathbb{E}[X]}{\sqrt{n\,\mathrm{Var}(X)}} \xrightarrow{d} \mathcal{N}(0,1)$$

meaning the CDF of the normalized empirical mean converges to the CDF of the standard normal distribution. Using this theorem we run the approximation using the binomial distribution with $p = 0.3$ and $n = 1000$. Figure 2 illustrates the approximation result.

1.2 Approximation quality

We want to estimate how accurate is the estimate:

$$F_n(t) = \frac{1}{n}\sum_{i=1}^{n} \mathbb{I}_{X_i \le t} = \frac{1}{n}\sum_{i=1}^{n} \mathbb{I}_{(-\infty,t]}(X_i)$$

The estimate is unbiased since:

$$\begin{aligned}\mathbb{E}\left[\frac{1}{n}\sum_{i=1}^{n} \mathbb{I}_{X_i \le t}\right] = \frac{1}{n}\sum_{i=1}^{n} \mathbb{E}\left[\mathbb{I}_{X_i \le t}\right] &\overset{*}{=} \frac{1}{n}\sum_{i=1}^{n} \mathbb{P}(X_i \le t) \\ &\overset{**}{=} \frac{1}{n}\sum_{i=1}^{n} \mathbb{P}(X \le t) = \mathbb{P}(X \le t)\end{aligned} \tag{1}$$

where $*$ holds since as we integrate the indicator function we get the probability of the event, and $**$ holds since $X_i \sim X$.

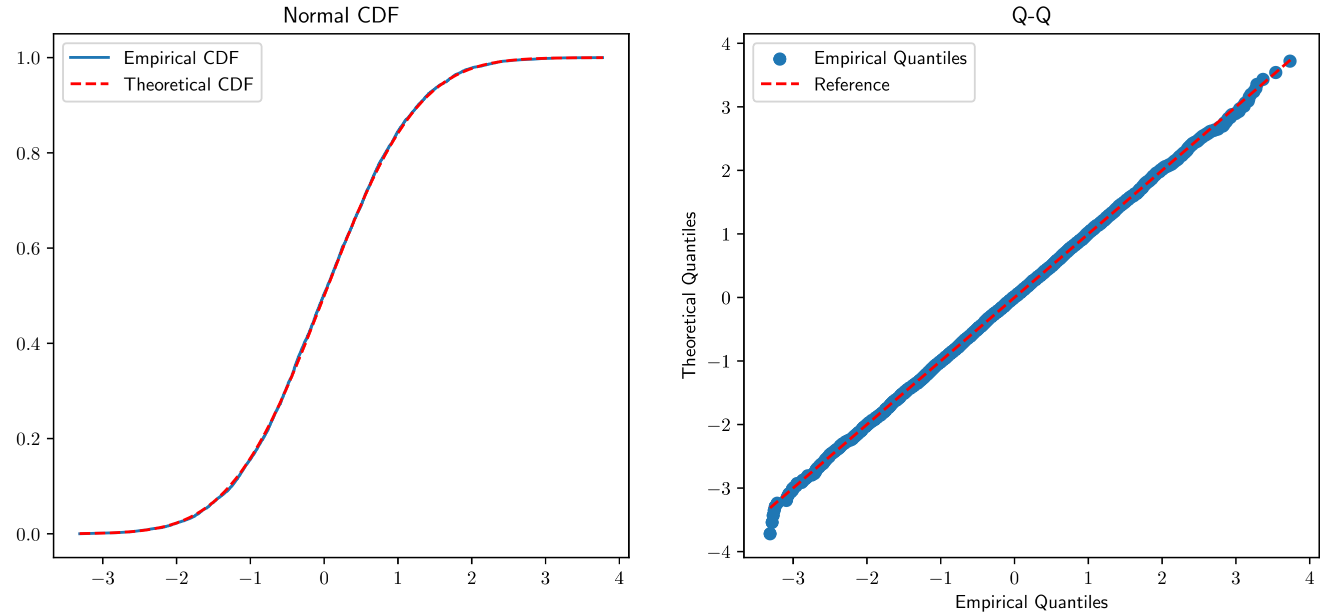

Figure 2: CDF and Q-Q plot

Each $\mathbb{I}_{X_i \leq t}$ is bounded by $[0,1]$, so we can use Hoeffding's inequality to get a bound on the probability of the estimate being far from the true value:

$$\mathbb{P}\left(\left|\frac{1}{n}\sum_{i=1}^{n}\mathbb{I}_{X_i \leq t} - \mathbb{P}(X \leq t)\right| \geq \delta\right) = \mathbb{P}\left(\left|\frac{1}{n}\sum_{i=1}^{n}\mathbb{I}_{X_i \leq t} - F(t)\right| \geq \delta\right) \leq 2e^{-2n\delta^2} \tag{2}$$

Figure 3 illustrates the bound for different values of δ.

For the almost sure convergence refer to Glivenko—Cantelli theorem, which states the following:

Theorem 1.1 (Glivenko—Cantelli theorem)**.** *Let $X_1, X_2, ... \overset{iid}{\sim} X$ and $F(x)$ be CDF of X. Moreover let F be continuous. Then we have:*

$$\mathbb{P}\left(\sup_{t \in \mathbb{R}} |F_n(t) - F(t)| \to 0\right) = 1$$

where

$$F_n(t) = \frac{1}{n}\sum_{i=1}^{n}\mathbb{I}_{(-\infty,t]}(X_i)$$

Proof of Theorem 1.1. Fix $-\infty = x_0 \leq x_1 \leq ... \leq x_m = \infty$, s.t. $F(x_k) + 1/m = F(x_{k+1})$ (using intermediate value theorem assuming F is continuous) for all $k \in \{0, ..., m-1\}$. Then if $x \in [x_{k-1}, x_k]$ by monotonicity of F_n and F we have:

- $F_n(x) - F(x) \geq F_n(x_{k-1}) - F(x_k) = F_n(x_{k-1}) - F(x_{k-1}) - 1/m$
- $F_n(x) - F(x) \leq F_n(x_k) - F(x_{k-1}) = F_n(x_k) - F(x_k) + 1/m$

Thus we have that for each $x \in \mathbb{R}$ there exists $k \in \{1, ..., m\}$ s.t. $x \in [x_{k-1}, x_k]$ hence:

$$\sup_{x \in \mathbb{R}} |F_n(x) - F(x)| \leq \max_{k \in \{0,...,m\}} |F_n(x_k) - F(x_k)| + 1/m \tag{3}$$

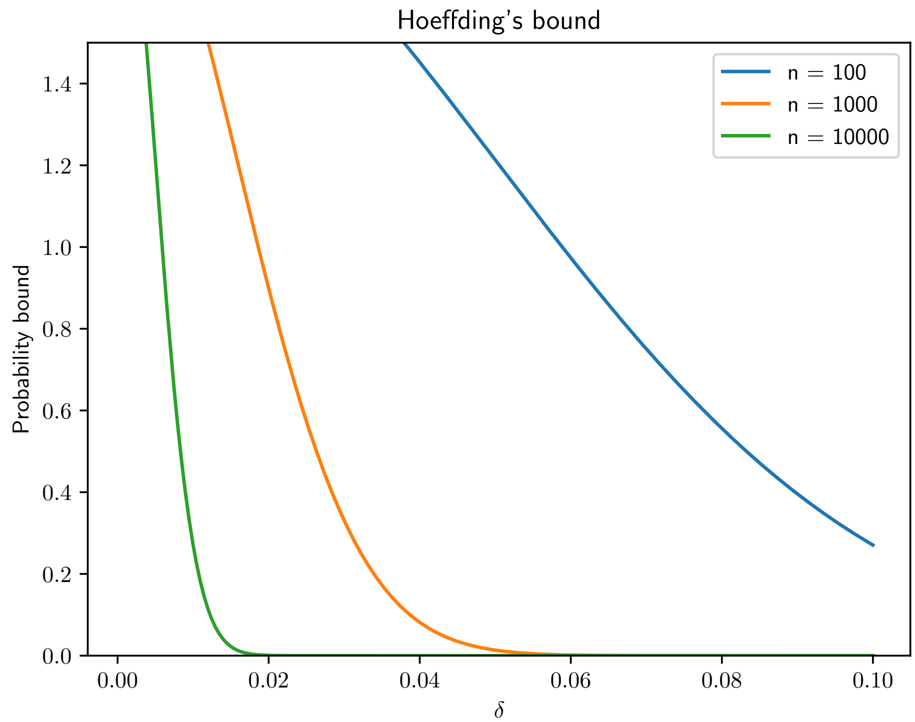

Figure 3: Hoeffding's inequality bound

If we now suppose that for some $\omega \in \Omega$:

$$\sup_{x\in\mathbb{R}} |F_n(x)(\omega) - F(x)| > \varepsilon + 1/m$$

then by (3) we have that:

$$\max_{k\in\{0,\dots,m\}} |F_n(x_k)(\omega) - F(x_k)| > \varepsilon$$

thus:

$$\mathbb{P}\left(\sup_{x\in\mathbb{R}} |F_n(x) - F(x)| > \varepsilon + 1/m \text{ i.o.}\right) \leq \mathbb{P}\left(\max_{k\in\{0,\dots,m\}} |F_n(x_k) - F(x_k)| > \varepsilon \text{ i.o.}\right) \tag{4}$$

as $\forall n \in \mathbb{N}$ for which the left hand side event occurs the right hand side event event occurs as well.

Since $F_n(t)$ is the sum of n independent random variables, we can use the strong law of large numbers (the estimate is unbiased (1), $\mathbb{I}_{(-\infty,t]}(X_i)$ are independent) to get that for each $k \in \{0, ..., m\}$:

$$\mathbb{P}\left(\lim_{n\to\infty} F_n(x_k) = F(x_k)\right) = 1$$

Thus for each $\varepsilon > 0$ we have:

$$\mathbb{P}\left(|F_n(x_k) - F(x_k)| > \varepsilon \text{ i.o.}\right) = 0$$

and by union bound we get:

$$\mathbb{P}\left(\bigcup_{k=0}^{m}\{|F_n(x_k) - F(x_k)| > \varepsilon \text{ i.o.}\}\right) \leq \sum_{k=0}^{m} \mathbb{P}\left(|F_n(x_k) - F(x_k)| > \varepsilon \text{ i.o.}\right) = 0$$

Claim 1. Let $(A_n)_{n\in\mathbb{N}}, (B_n)_{n\in\mathbb{N}} \in \mathcal{F}$ be two sequences of events. Then we have:

$$\limsup_{n\to\infty}(A_n \cup B_n) = \limsup_{n\to\infty} A_n \cup \limsup_{n\to\infty} B_n$$

Proof of Claim 1.

$$\limsup_{n\to\infty}(A_n \cup B_n) = \bigcap_{n=1}^{\infty}\bigcup_{m=n}^{\infty}(A_m \cup B_m) \stackrel{*}{=} \bigcap_{n=1}^{\infty}\left(\left(\bigcup_{m=n}^{\infty} A_m\right) \cup \left(\bigcup_{m=n}^{\infty} B_m\right)\right)$$

where $*$ holds by associativity and commutativity of union.

Denote:

$$\tilde{A}_n = \bigcup_{m=n}^{\infty} A_m \quad \tilde{B}_n = \bigcup_{m=n}^{\infty} B_m$$

We have that $\tilde{A}_n$ and $\tilde{B}_n$ are decreasing sequences of events.

"$\subset$" Suppose:

$$\omega \in \bigcap_{n=1}^{\infty}(\tilde{A}_n \cup \tilde{B}_n) \Rightarrow \omega \in \tilde{A}_n \cup \tilde{B}_n \ \forall n \in \mathbb{N}$$

meaning for each $n \in \mathbb{N}$ $\omega \in \tilde{A}_n$ or $\omega \in \tilde{B}_n$. Assume that for some $k \in \mathbb{N}$ $\omega \notin \tilde{A}_k$. Then for any $n > k$ we have that $\omega \notin \tilde{A}_n$, so we must have $\omega \in \tilde{B}_n$ for all $n > k$. Since $\tilde{B}_n$ is decreasing:

$$\omega \in \bigcap_{n=k}^{\infty} \tilde{B}_n = \bigcap_{n=1}^{\infty} \tilde{B}_n$$

Otherwise if $\omega \in \tilde{A}_n$ for all $n \in \mathbb{N}$ we have:

$$\omega \in \bigcap_{n=1}^{\infty} \tilde{A}_n$$

which implies:

$$\omega \in \left(\bigcap_{n=1}^{\infty} \tilde{A}_n\right) \cup \left(\bigcap_{n=1}^{\infty} \tilde{B}_n\right)$$

"$\supset$" Suppose:

$$\omega \in \left(\bigcap_{n=1}^{\infty} \tilde{A}_n\right) \cup \left(\bigcap_{n=1}^{\infty} \tilde{B}_n\right)$$

which means either $\omega \in \bigcap_{n=1}^{\infty} \tilde{A}_n$ or $\omega \in \bigcap_{n=1}^{\infty} \tilde{B}_n$. In the first case we have that $\omega \in \tilde{A}_n$ for all $n \in \mathbb{N}$, which implies $\omega \in \tilde{A}_n \cup \tilde{B}_n$ for all $n \in \mathbb{N}$. The second case is analogous.

□

With the Claim 1 we can rewrite the union of events as:

$$\left\{\max_{k\in\{0,\dots,m\}} |F_n(x_k)-F(x_k)| > \varepsilon \text{ i.o.}\right\} = \left\{\left(\bigcup_{k\in\{0,\dots,m\}} |F_n(x_k)-F(x_k)| > \varepsilon\right) \text{ i.o.}\right\}$$
$$= \left\{\bigcup_{k\in\{0,\dots,m\}} (|F_n(x_k)-F(x_k)| > \varepsilon \text{ i.o.})\right\} \tag{5}$$

Equation 4 together with equation 5 gives us:

$$\mathbb{P}\left(\sup_{x\in\mathbb{R}} |F_n(x)-F(x)| > \varepsilon + 1/m \text{ i.o.}\right) = 0$$

Finally since m is arbitrary we can choose $m \geq 2/\tilde{\varepsilon}$ and $\varepsilon = \tilde{\varepsilon}/2$ to get:

$$\mathbb{P}\left(\sup_{x\in\mathbb{R}} |F_n(x)-F(x)| > \tilde{\varepsilon} \text{ i.o.}\right) = 0 \quad \forall \tilde{\varepsilon} > 0$$

which is equivalent to:

$$\mathbb{P}\left(\sup_{x\in\mathbb{R}} |F_n(x)-F(x)| \to 0\right) = 1$$

□

Remark 1.1. The proof translates bounding an uncountable set of points to bounding a finite set of points. This result is of interest since it shows uniform convergence of the empirical CDF including the tails of the distribution, which is important for statistical inference.

Theorem 1.2 (Rate of convergence)**.** *Let $X_1, X_2, \dots \overset{iid}{\sim} X$ and $F(x)$ be CDF of X. Moreover let F be continuous. Then we have:*

$$\mathbb{P}\left(\sup_{t\in\mathbb{R}} |F_n(t)-F(t)| \geq \delta\right) \leq 4\left(1+\frac{1}{\delta}\right) e^{-\frac{n\delta^2}{2}}, \quad \forall \delta > 0$$

where

$$F_n(t) = \frac{1}{n}\sum_{i=1}^{n} \mathbb{I}_{(-\infty,t]}(X_i)$$

Proof of Theorem 1.2. Choose the points $x_0, \dots, x_m$ as in the proof of Theorem 1.1. Then using the bound (2) we have:

$$\mathbb{P}\left(\left|\frac{1}{n}\sum_{i=1}^{n} \mathbb{I}_{X_i\leq t} - F(t)\right| \geq \varepsilon\right) = \mathbb{P}\left(|F_n(t)-F(t)| \geq \varepsilon\right) \leq 2e^{-2n\varepsilon^2}$$

for all $t \in \mathbb{R}$ including the points $x_0, \dots, x_m$. Then using the union bound we have:

$$\mathbb{P}\left(\max_{k\in\{0,\dots,m\}} |F_n(x_k)-F(x_k)| \geq \varepsilon\right) \leq 2(m+1)e^{-2n\varepsilon^2}$$

Similarly to equation 4 we have that:

$$\mathbb{P}\left(\sup_{x\in\mathbb{R}}|F_n(x)-F(x)|\geq\varepsilon+1/m\right)$$
$$\leq\mathbb{P}\left(\max_{k\in\{0,\ldots,m\}}|F_n(x_k)-F(x_k)|\geq\varepsilon\right)\leq 2(m+1)e^{-2n\varepsilon^2}$$

which holds for any $m\in\mathbb{N}$.

Now fix $\delta>0$, choose $m=\lceil 2/\delta\rceil$, and substitute $\varepsilon=\delta/2$ into the inequality above:

$$\mathbb{P}\left(\sup_{x\in\mathbb{R}}|F_n(x)-F(x)|\geq\frac{\delta}{2}+\frac{1}{m}\right)\leq 2(m+1)e^{-\frac{n\delta^2}{2}}$$

Since $m=\lceil 2/\delta\rceil$, we have $1/m\leq\delta/2$, hence:

$$\left\{\sup_{x\in\mathbb{R}}|F_n(x)-F(x)|\geq\delta\right\}\subseteq\left\{\sup_{x\in\mathbb{R}}|F_n(x)-F(x)|\geq\frac{\delta}{2}+\frac{1}{m}\right\}$$

and therefore:

$$\mathbb{P}\left(\sup_{x\in\mathbb{R}}|F_n(x)-F(x)|\geq\delta\right)\leq 2(m+1)e^{-\frac{n\delta^2}{2}}$$

Finally, using $m=\lceil 2/\delta\rceil\leq 2/\delta+1$, we get:

$$2(m+1)\leq 2\left(\frac{2}{\delta}+2\right)=4\left(1+\frac{1}{\delta}\right)$$

so:

$$\mathbb{P}\left(\sup_{x\in\mathbb{R}}|F_n(x)-F(x)|\geq\delta\right)\leq 4\left(1+\frac{1}{\delta}\right)e^{-\frac{n\delta^2}{2}}$$

□

Remark 1.2. For the tighter bound refer to Dvoretzky—Kiefer—Wolfowitz inequality, which states the following:

$$\mathbb{P}\left(\sup_{t\in\mathbb{R}}|F_n(t)-F(t)|\geq\delta\right)\leq 2e^{-2n\delta^2},\quad\forall\delta\geq 0$$

Note that this is the same exact bound as in (2) but uniform over all $t\in\mathbb{R}$

Remark 1.3. By VC theory we have that the hypothesis class:

$$\mathcal{H}=\{h_t=\mathbb{I}_{(-\infty,t]}:t\in\mathbb{R}\}$$

has VC dimension $d=1$. Note that for the distribution $\mathcal{D}$ induced by $(X,0)$, where X is a random variable and 0 is the label, we have that the training loss for the sample $\mathcal{T}=((X_1,0),...,(X_n,0))\overset{iid}{\sim}\mathcal{D}$ is given by:

$$\mathscr{L}_{\mathcal{T}}=\frac{1}{n}\sum_{i=1}^{n}\ell_{0-1}(\mathbb{I}_{(-\infty,t]}(X_i),0)=\frac{1}{n}\sum_{i=1}^{n}\mathbb{I}_{(-\infty,t]}(X_i)=F_n(t)$$

The true loss is given by:

$$\mathscr{L}_{\mathcal{D}}(h_t)=\mathbb{E}_{(X,0)\sim\mathcal{D}}[\ell_{0-1}(\mathbb{I}_{(-\infty,t]}(X),0)]=\mathbb{E}_{X\sim\mathcal{D}}[\mathbb{I}_{(-\infty,t]}(X)]=\mathbb{P}(X\leq t)=F(t)$$

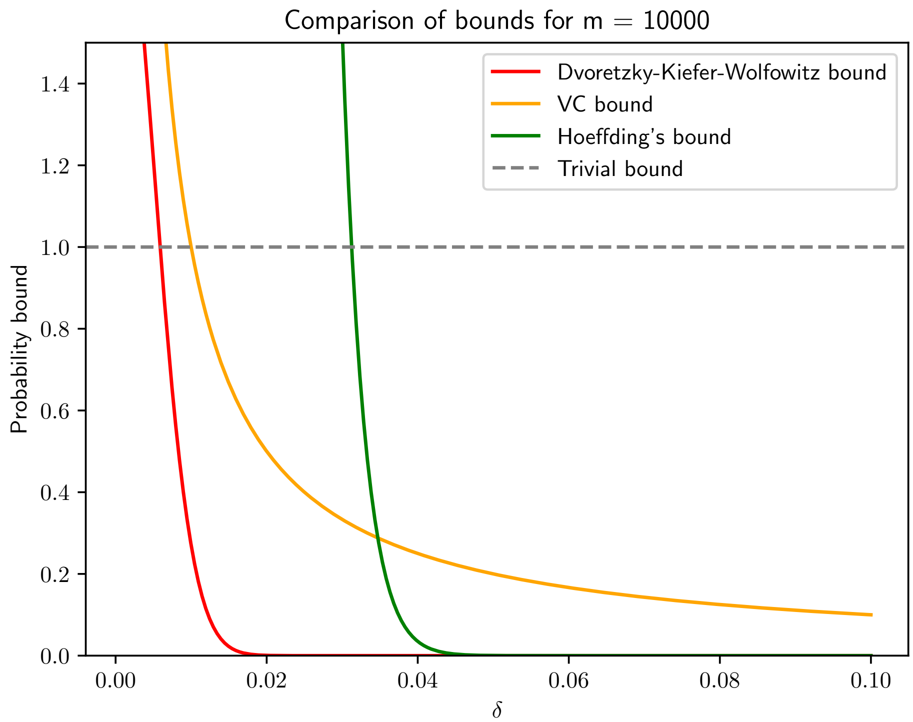

Figure 4: Bounds comparison: Hoeffding's inequality (Theorem 1.2), Dvoretzky—Kiefer—Wolfowitz inequality (Remark 1.2), VC theory bound (Remark 1.3)

By Fundamental Theorem of Statistical Learning Theory for any distribution $\mathcal{D}$:

$$\mathbb{E}\left[\sup_{h\in\mathcal{H}}|\mathscr{L}_{\mathcal{T}}(h)-\mathscr{L}_{\mathcal{D}}(h)|\right]=\mathbb{E}\left[\sup_{t\in\mathbb{R}}|F_n(t)-F(t)|\right]\leq C\sqrt{\frac{\mathrm{VC}(\mathcal{H})}{n}}$$

for some constant $C>0$.

By Markov's inequality we have that for any $\delta>0$:

$$\mathbb{P}\left(\sup_{t\in\mathbb{R}}|F_n(t)-F(t)|\geq\delta\right)\leq\frac{\mathbb{E}\left[\sup_{t\in\mathbb{R}}|F_n(t)-F(t)|\right]}{\delta}\leq\frac{C\sqrt{\frac{\mathrm{VC}(\mathcal{H})}{n}}}{\delta}=C\sqrt{\frac{1}{n\delta^2}}$$

2 Linear regression

Assume the following construction:

$$y_i = \beta^T x_i + \varepsilon_i, \quad \varepsilon_1, ..., \varepsilon_n \overset{iid.}{\sim} \mathcal{N}(0, \sigma^2)$$

where $x_i \in \mathbb{R}^d$ and $\beta \in \mathbb{R}^d$ are fixed

2.1 Preliminaries

Lemma 2.1. *Let $X : \Omega \to \mathbb{R}^d$ be a random variable and $A \in \mathbb{R}^{d \times d}, b \in \mathbb{R}^d$. Assume that the second moments exist, i.e. $\mathbb{E}[X_i^2] < \infty$ and the covariance matrix is given by Σ. Then the following properties hold:*

Property 2.1.1 $\mathbb{E}[AX + b] = A \cdot \mathbb{E}[X] + b$

Property 2.1.2 $\mathrm{Cov}(AX) = A \cdot \Sigma \cdot A^T$

Remark 2.1. Using the fact that the second moment exists and by Jensen's inequality, we have that $\mathbb{E}[X_i] < \infty$ and $\mathbb{E}[X_i^2] < \infty$ for all $i = 1, ..., d$. Therefore the covariance matrix is well-defined, since $(\mathbb{E}[|X_i X_j|])^2 = (\mathbb{E}[|X_i||X_j|])^2 \leq \mathbb{E}[|X_i|^2]\,\mathbb{E}[|X_j|^2] < \infty$ by Cauchy-Schwarz inequality.

Proof.

1. Trivial

2. Suppose that the expected value of X is the zero vector. By Property 2.1.1 we also have that $\mathbb{E}[AX] = A\,\mathbb{E}[X] = 0$. Then we have that:

$$\begin{aligned}
\mathrm{Cov}((AX)_i, (AX)_j) &= \mathrm{Cov}\left(\sum_{k=1}^d A_{ik}X_k, \sum_{l=1}^d A_{jl}X_l\right) = \mathbb{E}\left(\left(\sum_{k=1}^d A_{ik}X_k\right)\left(\sum_{l=1}^d A_{jl}X_l\right)\right) \\
&= \mathbb{E}\left[\sum_{k=1}^d \sum_{l=1}^d A_{ik}X_kX_lA_{lj}^T\right] = \sum_{k=1}^d \sum_{l=1}^d A_{ik}\,\mathbb{E}[X_kX_l]A_{lj}^T \\
&= \sum_{k=1}^d A_{ik}\left(\sum_{l=1}^d \Sigma_{kl}A_{lj}^T\right) = \sum_{k=1}^d A_{ik}(\Sigma A^T)_{kj} = (A\Sigma A^T)_{ij}
\end{aligned}$$

If the expected value of X is not the zero vector, we can define a new random variable $Y = X - \mathbb{E}[X]$, which has the expected value of zero. Then we have that:

$$\mathrm{Cov}(Y_i, Y_j) = \mathbb{E}[Y_iY_j] = \mathbb{E}[(X_i - \mathbb{E}[X]_i)(X_j - \mathbb{E}[X]_j)] = \mathrm{Cov}(X_i, X_j)$$

Therefore we have that:

$$\begin{aligned}
\mathrm{Cov}((AX)_i, (AX)_j) &= \mathbb{E}[(AX - \mathbb{E}[AX])_i \cdot (AX - \mathbb{E}[AX])_j] \\
&\overset{*}{=} \mathbb{E}[(A(X - \mathbb{E}[X]))_i \cdot (A(X - \mathbb{E}[X]))_j] \\
&= \mathbb{E}[(AY)_i \cdot (AY)_j] = \mathrm{Cov}((AY)_i, (AY)_j) \\
&= (A\,\mathrm{Cov}(Y)A^T)_{ij} = (A\,\mathrm{Cov}(X)A^T)_{ij}
\end{aligned}$$

where $*$ follows from Property 2.1.1.

□

Lemma 2.2 (Multivariate normal distribution)**.** *We say that* $X : \Omega \to \mathbb{R}^d$ *is multivariately normally distributed if and only if there exists a matrix* $A \in \mathbb{R}^{d\times d}$ *and a vector* $b \in \mathbb{R}^d$ *such that* $X = AZ + b$*, where* $Z_1, ..., Z_d \overset{iid.}{\sim} \mathcal{N}(0, 1)$*.*

We have that multivariate normal distribution is closed under linear transformations, i.e. if $X \sim \mathcal{N}(\mu, \Sigma)$ *and* $Y = AX + b$*, then* $Y \sim \mathcal{N}(A\mu + b, A\Sigma A^T)$*. (For the expectation and covariance refer to Property 2.1.1 and Property 2.1.2).*

Lemma 2.3. *For the multivariate normal distribution, we have that the components are independent if and only if they are uncorrelated, i.e.* $\mathrm{Cov}(X_i, X_j) = 0$ *for all* $i \neq j$*.*

Lemma 2.4. *Let* $X \in \mathbb{R}^{n\times d}$ *be a matrix of full rank, where* $n > d$*. Define* $H = X(X^TX)^{-1}X^T \in \mathbb{R}^{n\times n}$ *and* $E = \mathrm{diag}\{\underbrace{1, ..., 1}_{n-d}, \underbrace{0, ..., 0}_{d}\}$*, then:*

Property 2.4.1 $I - H$ *is symmetric and idempotent.*

Property 2.4.2 $\mathrm{rank}(I - H) = n - d$

Property 2.4.3 There exists a matrix Q *such that* $I - H = Q^TEQ$

Property 2.4.4 $\ker(I - H) = \mathrm{im}(X)$

Proof.

1. This follows by the direct computation:

$$(I - H)^T = I - H^T = I - (X(X^TX)^{-1}X^T)^T = I - (X^T)^T \left((X^TX)^T\right)^{-1} X^T = I - H$$

$$\begin{aligned}(I - H)(I - H) &= (I - X(X^TX)^{-1}X^T)(I - X(X^TX)^{-1}X^T)\\ &= I - 2X(X^TX)^{-1}X^T + X(X^TX)^{-1}X^TX(X^TX)^{-1}X^T\\ &= I - 2X(X^TX)^{-1}X^T + X(X^TX)^{-1}X^T\\ &= I - X(X^TX)^{-1}X^T = I - H\end{aligned}$$

2. Since $I - H$ is idempotent we have that its trace is equal to its rank. Therefore we have that:

$$\mathrm{trace}(H) = \mathrm{trace}(X(X^TX)^{-1}X^T) \overset{*}{=} \mathrm{trace}((X^TX)^{-1}X^TX) = \mathrm{trace}(I_d) = d$$

where $*$ follows from the cyclic property of the trace. Therefore we have that:

$$\mathrm{rank}(I - H) = \mathrm{trace}(I - H) = \mathrm{trace}(I) - \mathrm{trace}(H) = n - d$$

3. Since the matrix $I - H$ is idempotent we have that it has an eigenvalue 1 with multiplicity equal to $\mathrm{rank}(I - H) = n - d$ and eigenvalue 0 with multiplicity d. Its minimal polynomial is contained in $\{x, x - 1, x(x - 1)\}$ implying that $I - H$ is diagonalizable since in all the cases it does not have repeated factors. Therefore there exists a matrix $Q \in O(\mathbb{R}^n)$, s.t.

$$Q^T(I - H)Q = E$$

4. By the rank-nullity theorem we have $\dim\ker(I-H) = n - \operatorname{rank}(I-H) = n-(n-d) = d$. Using the assumption we have that $\dim\operatorname{im}(X) = d$. Therefore it suffices to show that $\operatorname{im} X \subset \ker(I-H)$:

$$\begin{aligned}(I-H)Xy &= (I - X(X^TX)^{-1}X^T)Xy \\ &= Xy - X(X^TX)^{-1}X^TXy = Xy - Xy = 0 \quad \forall y \in \mathbb{R}^d\end{aligned}$$

This shows that $\operatorname{im} X \subset \ker(I-H)$, and since they have the same dimension we have that $\operatorname{im} X = \ker(I-H)$.

□

2.2 t-statitistic

In this section we assume that $\hat{\beta}$ is the ordinary least squares estimator of β. We establish two facts:

- $\varepsilon \sim \mathcal{N}(0, \sigma^2 I)$ since the components of ε are independent and identically distributed as $\mathcal{N}(0,\sigma^2)$, thus we can obtain $\varepsilon = \sigma I \cdot (\varepsilon/\sigma)$, where $\varepsilon_1/\sigma, ..., \varepsilon_n/\sigma \overset{iid.}{\sim} \mathcal{N}(0,1)$.
- $y = X\beta + \varepsilon \sim \mathcal{N}(X\beta, \sigma^2 I)$ as a consequence of Lemma 2.2 and the fact that y is a linear transformation of ε.

Step 1. Compute the distribution of the residual sum of squares:

$$\hat{\varepsilon} = y - X\hat{\beta} = y - X(X^TX)^{-1}X^Ty = (I-H)y$$

By Property 2.4.4 we have that $\operatorname{im}(X) = \ker(I-H)$, and since $X\beta \in \operatorname{im}(X)$ we have that:

$$(I-H)X\beta = 0$$

thus:

$$(I-H)y = (I-H)(X\beta + \varepsilon) = (I-H)\varepsilon$$

with the notation from Lemma 2.4. Using the result from Property 2.4.3 we have that:

$$I - H = Q^TEQ$$

for some orthogonal matrix Q.

Writing out the expression for the residual sum of squares, we have that:

$$\text{RSS} = \hat{\varepsilon}^T\hat{\varepsilon} = \varepsilon^T(I-H)^T(I-H)\varepsilon = \varepsilon^T(I-H)\varepsilon$$

Define $U = Q\varepsilon$. Then $U \sim \mathcal{N}(0,\sigma^2 I)$ by Lemma 2.2. Then we have that:

- $\mathbb{E}[Q\varepsilon] = Q\,\mathbb{E}[\varepsilon] = 0$ by Property 2.1.1
- $\operatorname{Cov}(Q\varepsilon) = Q\operatorname{Cov}(\varepsilon)Q^T = Q\sigma^2 IQ^T = \sigma^2 QQ^T = \sigma^2 I$ by Property 2.1.2

Then by Lemma 2.3 we have that the components of U are independent, since they are uncorrelated.

Thus we have that U/σ follows a joint standard normal distribution. Then we have that:

$$\mathrm{RSS} = U^T E U \tag{6}$$

$$\frac{\mathrm{RSS}}{\sigma^2} = U^T/\sigma \cdot E \cdot U/\sigma = \sum_{i=1}^{n-d} (U_i/\sigma)^2 \sim \chi^2_{n-d}$$

Step 2. Compute the distribution of $\hat{\beta}_i - \beta_i$:

Observe that the $\hat{\beta}$ follows multivariate normal distribution since $\hat{\beta} = Ay$ where $A = (X^T X)^{-1} X^T$, and $y \sim \mathcal{N}(X\beta, \sigma^2 I)$ (see Lemma 2.2). Therefore $\hat{\beta} \sim \mathcal{N}(\beta, \sigma^2 AA^T)$, where we have used that $\hat{\beta}$ is an unbiased estimator of β and that

$$\mathrm{Cov}(\hat{\beta}) = \mathrm{Cov}(Ay) = A\,\mathrm{Cov}(y)A^T = \sigma^2 AA^T$$

by Property 2.1.2.

$$AA^T = (X^T X)^{-1} X^T X \left((X^T X)^{-1}\right)^T = \left((X^T X)^{-1}\right)^T = \left((X^T X)^T\right)^{-1} = (X^T X)^{-1}$$

As was derived above, we have that $\hat{\beta} \sim \mathcal{N}(\beta, \sigma^2 (X^T X)^{-1})$. Therefore we have that:

$$\hat{\beta}_i - \beta_i \sim \mathcal{N}(0, \sigma^2 (X^T X)^{-1}_{ii})$$

Thus:

$$\frac{\hat{\beta}_i - \beta_i}{\sigma\sqrt{(X^T X)^{-1}_{ii}}} \sim \mathcal{N}(0,1)$$

Step 3. $\hat{\varepsilon}$ and $\hat{\beta} - \beta$ are independent:

We write $\hat{\varepsilon} = (I - H)\varepsilon$ and $\hat{\beta} - \beta = (X^T X)^{-1} X^T (X\beta + \varepsilon) - \beta = (X^T X)^{-1} X^T \varepsilon$. Then we have that:

$$\begin{pmatrix} \hat{\varepsilon} \\ \hat{\beta} - \beta \end{pmatrix} = \begin{pmatrix} I - H \\ (X^T X)^{-1} X^T \end{pmatrix} \varepsilon$$

Thus by Lemma 2.2 we have that the joint distribution of $\hat{\varepsilon}$ and $\hat{\beta} - \beta$ is multivariate normal. Therefore by Lemma 2.3 it suffices to show that they are uncorrelated, i.e. $\mathrm{Cov}(\hat{\varepsilon}, \hat{\beta} - \beta) = 0$. We have that:

$$\begin{aligned} \mathrm{Cov}(\hat{\varepsilon}, \hat{\beta} - \beta) &\overset{*}{=} \mathbb{E}[\hat{\varepsilon}(\hat{\beta} - \beta)^T] = \mathbb{E}[(I - H)\varepsilon\varepsilon^T X (X^T X)^{-1}] \\ &\overset{**}{=} (I - H)\,\mathbb{E}[\varepsilon\varepsilon^T] X (X^T X)^{-1} \\ &= \sigma^2 (I - H) X (X^T X)^{-1} \end{aligned}$$

where $*$ follows from the fact that both $\hat{\varepsilon}$ and $\hat{\beta} - \beta$ have expected value of zero as the linear transformation of ε which has expected value of zero. $**$ follows from Property 2.1.1.

By Lemma Property 2.4.4 we have that $\mathrm{im}(X) = \ker(I - H)$, thus $(I - H)X = 0$, hence:

$$\mathrm{Cov}(\hat{\varepsilon}, \hat{\beta} - \beta) = \sigma^2 (I - H) X (X^T X)^{-1} = 0$$

Step 4. Conclude that the t-statistic follows a t-distribution with $n-d$ degrees of freedom:

$$\frac{\hat{\beta}_i - \beta_i}{\sqrt{\frac{\text{RSS}}{n-d}(X^TX)^{-1}_{ii}}} = \frac{\frac{\hat{\beta}_i-\beta_i}{\sigma\sqrt{(X^TX)^{-1}_{ii}}}}{\frac{1}{\sigma}\sqrt{\frac{\text{RSS}}{n-d}}} = \frac{\frac{\hat{\beta}_i-\beta_i}{\sigma\sqrt{(X^TX)^{-1}_{ii}}}}{\sqrt{\frac{\text{RSS}}{\sigma^2(n-d)}}} = \frac{Z}{\sqrt{\frac{T}{n-d}}}$$

where by the previous steps we have that $Z = \frac{\hat{\beta}_i-\beta_i}{\sigma\sqrt{(X^TX)^{-1}_{ii}}} \sim \mathcal{N}(0,1)$ and $T = \frac{\text{RSS}}{\sigma^2} \sim \chi^2_{n-d}$. T depends only on the residuals and Z depends only on $\hat{\beta}_i$, which are independent by the previous step. Therefore we have that the t-statistic follows a t-distribution with $n-d$ degrees of freedom.

Theorem 2.1. *The unbiased estimator of the variance of the error term is given by:*

$$\hat{\sigma}^2 = \frac{1}{n-d}\sum_{i=1}^{n} \hat{\varepsilon}_i^2$$

where d is the number of features and n is the number of samples.

Proof.

$$\text{RSS} = \hat{\varepsilon}^T\hat{\varepsilon} \overset{(6)}{=} \sum_{i=1}^{n-d} U_i^2$$

Each U_i is distributed as $\mathcal{N}(0,\sigma^2)$, therefore we have that:

$$\mathbb{E}[U_i^2] = \text{Var}(U_i) = \sigma^2$$

Thus we have that:

$$\mathbb{E}[\text{RSS}] = \mathbb{E}\left[\sum_{i=1}^{n-d} U_i^2\right] = \sum_{i=1}^{n-d}\mathbb{E}[U_i^2] = (n-d)\sigma^2$$

Dividing both sides by $n-d$ yields:

$$\mathbb{E}\left[\frac{\text{RSS}}{n-d}\right] = \sigma^2$$

□

Remark 2.2. This provides an intuition why we use this estimator for the variance of the error term.

Theorem 2.2 (Gauss-Markov theorem)**.** *The ordinary least squares estimator $\hat{\beta}$ is the best linear unbiased estimator (BLUE) of β, i.e. for each $C \in \mathbb{R}^{d\times n}$ such that $\mathbb{E}[Cy] = \beta$, we have that:*

$$\text{Cov}(\hat{\beta}) \preceq \text{Cov}(Cy)$$

Proof. We need to show that the matrix:

$$\mathrm{Cov}(Cy) - \mathrm{Cov}(\hat{\beta})$$

is positive semidefinite.

Define $D = C - (X^T X)^{-1} X^T$. Then we have that:

$$\mathbb{E}[Cy] = \mathbb{E}[((X^T X)^{-1} X^T + D)y] = \mathbb{E}[(X^T X)^{-1} X^T y] + \mathbb{E}[Dy] = \beta$$

Thus we have that $\mathbb{E}[Dy] = 0$. Therefore we have that:

$$\mathbb{E}[Dy] = \mathbb{E}[DX\beta + D\varepsilon] = DX\beta = 0$$

Since β is unknown we must have that $DX\beta = 0$ for all β, which implies that $DX = 0$. This also implies $(DX)^T = X^T D^T = 0$. Then we have that:

$$\begin{aligned}
\mathrm{Cov}(Cy) &= \sigma^2 CC^T \\
&= \sigma^2((X^T X)^{-1} X^T + D)((X^T X)^{-1} X^T + D)^T \\
&= \sigma^2((X^T X)^{-1} X^T X (X^T X)^{-1} + DX(X^T X)^{-1} + (X^T X)^{-1} X^T D^T + DD^T) \\
&= \sigma^2((X^T X)^{-1} + DD^T)
\end{aligned}$$

We know that DD^T is positive semidefinite, so for any $v \in \mathbb{R}^d$ we have that:

$$v^T CC^T v = v^T (X^T X)^{-1} v + v^T DD^T v \geq v^T (X^T X)^{-1} v$$

thus:

$$v^T (CC^T - (X^T X)^{-1}) v \geq 0$$

implying that $CC^T - (X^T X)^{-1}$ is positive semidefinite. Therefore we have that:

$$\mathrm{Cov}(Cy) - \mathrm{Cov}(\hat{\beta}) = \sigma^2(CC^T - (X^T X)^{-1}) \succeq 0$$

□

3 Box-Muller transformation

Consider a random variable $Z \sim \mathcal{N}(0, I_2)$, meaning $f_Z = \frac{1}{2\pi} e^{-(x^2+y^2)/2}$. Let $z, t \in \mathbb{R}$ be arbitrary. Then we have

$$\begin{aligned} \mathbb{P}(Z \in (-\infty, z] \times (-\infty, t]) &= \int_{(-\infty,z]\times(-\infty,t]} f_Z(x,y)\, d(x,y) \\ &= \int_{\mathbb{R}^2} \frac{1}{2\pi} e^{-(x^2+y^2)/2} \cdot \mathbb{I}_{(-\infty,z]\times(-\infty,t]}(x,y)\, d(x,y) \end{aligned} \tag{7}$$

We do the change of variables:

$$\Phi : (0,1)^2 \to \mathbb{R}^2 \setminus \{(u,0) : u \geq 0\}, \quad (x,y) \mapsto \begin{pmatrix} \sqrt{-2\ln x}\cos(2\pi y) \\ \sqrt{-2\ln x}\sin(2\pi y) \end{pmatrix} = \begin{pmatrix} \phi_1(x,y) \\ \phi_2(x,y) \end{pmatrix}$$

Claim 2. The map Φ is a bijection.

Proof of claim 2.

Surjectivity: Let $(z,t) \in \mathbb{R}^2 \setminus \{(u,0) : u \geq 0\}$.By definition of atan2 we have that:

$$\begin{pmatrix} z \\ t \end{pmatrix} = \sqrt{z^2+t^2} \begin{pmatrix} \cos(\operatorname{atan2}(t,z)) \\ \sin(\operatorname{atan2}(t,z)) \end{pmatrix}$$

for all $(z,t) \in \mathbb{R}^2$. atan2 $\in (-\pi, \pi]$, so we define $\theta = \operatorname{atan2}(-t,-z) + \pi \in (0, 2\pi]$. By the previous equation we have that:

$$\begin{pmatrix} \cos(\operatorname{atan2}(-t,-z)+\pi) \\ \sin(\operatorname{atan2}(-t,-z)+\pi) \end{pmatrix} = \begin{pmatrix} -\cos(\operatorname{atan2}(-t,-z)) \\ -\sin(\operatorname{atan2}(-t,-z)) \end{pmatrix} = \frac{1}{\sqrt{z^2+t^2}} \begin{pmatrix} z \\ t \end{pmatrix} \tag{8}$$

Since the ray $\{(u,0) : u \geq 0\}$ is excluded from the codomain of Φ, we have that $\theta \in (0, 2\pi)$, so we can define $y = \theta/(2\pi) \in (0,1)$. We also define $x = e^{-(z^2+t^2)/2} \in (0,1)$ (either z or t is non-zero) to obtain that:

$$\begin{aligned} \Phi(x,y) = \begin{pmatrix} \sqrt{-2\ln x}\cos(2\pi y) \\ \sqrt{-2\ln x}\sin(2\pi y) \end{pmatrix} &= \begin{pmatrix} \sqrt{-2(-(z^2+t^2)/2)}\cos(2\pi \cdot \theta/(2\pi)) \\ \sqrt{-2(-(z^2+t^2)/2)}\sin(2\pi \cdot \theta/(2\pi)) \end{pmatrix} \\ &= \sqrt{z^2+t^2} \begin{pmatrix} \cos(\theta) \\ \sin(\theta) \end{pmatrix} \overset{(8)}{=} \begin{pmatrix} z \\ t \end{pmatrix} \end{aligned}$$

Injectivity: Let $(x_1,y_1), (x_2,y_2) \in (0,1)^2$ such that $\Phi(x_1,y_1) = \Phi(x_2,y_2)$. Then we have that:

$$\sqrt{-2\ln x_1}\cos(2\pi y_1) = \sqrt{-2\ln x_2}\cos(2\pi y_2)$$

$$\sqrt{-2\ln x_1}\sin(2\pi y_1) = \sqrt{-2\ln x_2}\sin(2\pi y_2)$$

Squaring and adding the two equations gives us that $-2\ln x_1 = -2\ln x_2$, so $x_1 = x_2$. Dividing both sides by $\sqrt{-2\ln x_1}$ (which is non-zero since $x_1 \neq 1$) gives us that $\cos(2\pi y_1) = \cos(2\pi y_2)$ and $\sin(2\pi y_1) = \sin(2\pi y_2)$, which implies that $y_1 = y_2$ as $y_i \in (0,1)$.

$\square$

Now we compute the Jacobian of Φ:

$$\sqrt{-2\ln x}' = -\frac{2\ln'(x)}{2\sqrt{-2\ln x}} = -\frac{1}{x\sqrt{-2\ln x}}$$

Thus:

$$\mathrm{J}\,\Phi(x,y) = \begin{pmatrix} -\frac{1}{x\sqrt{-2\ln x}}\cos(2\pi y) & -2\pi\sqrt{-2\ln x}\sin(2\pi y) \\ -\frac{1}{x\sqrt{-2\ln x}}\sin(2\pi y) & 2\pi\sqrt{-2\ln x}\cos(2\pi y) \end{pmatrix}$$

The determinant of the Jacobian is:

$$\det \mathrm{J}\,\Phi(x,y) = -\frac{1}{x\sqrt{-2\ln x}} \cdot 2\pi\sqrt{-2\ln x} = -\frac{2\pi}{x}$$

where we used the trigonometric identity $\cos^2(2\pi y) + \sin^2(2\pi y) = 1$. Each component of the Jacobian is continuous on $(0,1)^2$, so Φ is continuously differentiable on $(0,1)^2$. The determinant of the Jacobian is non-zero on $(0,1)^2$, so Φ is a diffeomorphism.

We start plugging in the change of variables into (7):

$$x^2 + y^2 \xrightarrow{\Phi} -2\ln x$$

$$\frac{1}{2\pi}e^{-(x^2+y^2)/2} \xrightarrow{\Phi} \frac{1}{2\pi}e^{\ln(x)} = \frac{x}{2\pi}$$

$$\frac{1}{2\pi}e^{-(x^2+y^2)/2} \cdot |\det \mathrm{J}\,\Phi(x,y)| \xrightarrow{\Phi} \frac{x}{2\pi} \cdot \left|-\frac{2\pi}{x}\right| = \frac{x}{2\pi}\frac{2\pi}{x} = 1$$

as $x \in (0,1)$ if we restrict to the domain of Φ. Thus the integral in (7) becomes:

$$\begin{aligned}\int_{(-\infty,z]\times(-\infty,t]} f_Z(x,y)\,d(x,y) &= \int_{(0,1)^2} \mathbb{I}_{(-\infty,z]\times(-\infty,t]}(\Phi(x,y))\,d(x,y) \\ &= \int_{\mathbb{R}^2\setminus\{(u,0):u\geq 0\}} \mathbb{I}_{(-\infty,z]\times(-\infty,t]}(\Phi(x,y)) \cdot \mathbb{I}_{(0,1)^2}\ d(x,y) \\ &\overset{*}{=} \int_{\mathbb{R}^2} \mathbb{I}_{(-\infty,z]\times(-\infty,t]}(\Phi(x,y)) \cdot \mathbb{I}_{[0,1]^2}\ d(x,y) \end{aligned} \tag{9}$$

where in $*$ we used that $\lambda^2(\{(u,0) : u \geq 0\}) = 0$ as:

$$\begin{aligned}\{(u,0) : u \geq 0\} = \bigcup_{n\in\mathbb{N}} [0,n] \times \{0\} \Rightarrow \lambda^2(\{(u,0) : u \geq 0\}) \\ \leq \sum_{n=1}^{\infty} \lambda^2([0,n]\times\{0\}) = \sum_{n=1}^{\infty} n \cdot 0 = 0\end{aligned}$$

and that $\mathbb{I}_{(0,1)^2} = \mathbb{I}_{[0,1]^2}$ almost everywhere. Observe that $\mathbb{I}_{[0,1]^2}$ is the density of the uniform distribution on $[0,1]^2$, say $U = (U_1, U_2)$. Thus by the change of variables formula (10) and the fact that $\mathbb{P}(A) = \mathbb{E}[\mathbb{I}_A]$ we have that:

$$\mathbb{E}[f(X)] = \int_{\mathbb{R}^m} f(x) f_X(x)\,d(x), \quad X : \Omega \to \mathbb{R}^m, f : \mathbb{R}^m \to \mathbb{R} \tag{10}$$

$$\mathbb{P}(\Phi(U_1,U_2) \in (-\infty,z]\times(-\infty,t]) = \mathbb{E}[\mathbb{I}_{(-\infty,z]\times(-\infty,t]}(\Phi(U_1,U_2))] = (9)$$

Thus we have shown that:

$$\mathbb{P}(Z \in (-\infty,z]\times(-\infty,t]) = \mathbb{P}(\Phi(U_1,U_2) \in (-\infty,z]\times(-\infty,t])$$

so we can conclude that $\mathbb{P}_Z = \mathbb{P}_{\Phi(U)}$ by the uniqueness of measure as the intervals $(-\infty,z]\times(-\infty,t]$ generate the Borel σ-algebra on $\mathbb{R}^2$ and form a π-system.

4 Heap asymptotics

We heapify the container by levels starting from the bottom one. Let n be the Heap size. Then $(n-1)/2$ is the index of the first node having left child. As we assume that all the substrees starting from the level $i+1$ are valid min heaps, we have that the tree with the root at index j at the level i may only have the root out of order, which is fixed by sinking it down. The number of executions of the sinking down is given by:

$$\sum_{i=0}^{h-1}(2^i \cdot (h-i))$$

where h is the height of the heap, 2^i is the number of nodes at the level i and $h-i$ is the number of level below the i-th one.

Lemma 4.1. *For any $q \in (-1,1)$:*

$$\sum_{k=0}^{\infty} kq^k = \frac{q}{(1-q)^2}$$

Proof. Suppose $\sum_{k=1}^{\infty} kq^k$ converges for $|q|<1$. Let $S_n = \sum_{k=1}^{n} kq^k$:

$$\begin{aligned}
S_n - qS_n = \sum_{k=1}^{n} kq^k - \sum_{k=1}^{n} kq^{k+1} &= \sum_{k=1}^{n} kq^k - \sum_{k=2}^{n+1}(k-1)q^k \\
&= q + \sum_{k=2}^{n} kq^k - \sum_{k=2}^{n}(k-1)q^k - nq^{n+1} \\
&= q + \sum_{k=2}^{n}(k-k+1)q^k - nq^{n+1} \\
&= \sum_{k=1}^{n} q^k - nq^{n+1}
\end{aligned}$$

Since we assume the sequence S_n to converge, $nq^{n+1} \to 0$ as a sequence must converge to zero to be summable.

$$\begin{aligned}
(1-q)\lim_{n\to\infty} S_n = \lim_{n\to\infty}(S_n - qS_n) &= \lim_{n\to\infty}\left(\sum_{k=1}^{n} q^k - nq^{n+1}\right) \\
&= \frac{q}{1-q} - \lim_{n\to\infty} nq^{n+1} \\
&= \frac{q}{1-q}
\end{aligned}$$

Therefore:

$$\sum_{k=1}^{\infty} kq^k = \frac{q}{(1-q)^2}$$

Consequently:

$$\sum_{k=0}^{\infty} kq^k = 0 \cdot q^0 + \sum_{k=1}^{\infty} kq^k = \frac{q}{(1-q)^2}$$

To prove convergence of S_n we can use the ratio test:

$$\left|\frac{(k+1)q^{k+1}}{kq^k}\right| = |q|\frac{k+1}{k}$$

The sequence $(k+1)/k \to 1$. Since $|q| < 1$ we can pick θ in such a way that $|q| < \theta < 1$. Hence we have that there exists $n_0 \in \mathbb{N}$, s.t. $|(k+1)/k - 1| = (k+1)/k - 1 < 1/\theta - 1$ for all $n > n_0$

$$|q|\frac{k+1}{k} \le \frac{|q|}{\theta} < 1 \quad \forall n \ge n_0$$

$\square$

By substituting $k = h - i$ we get that the sum is equal to:

$$\begin{aligned}\sum_{i=0}^{h-1}(2^i \cdot (h-i)) = \sum_{k=1}^{h}(2^{h-k} \cdot k) = 2^h \sum_{k=1}^{h} 2^{-k} \cdot k < 2^h \sum_{k=0}^{\infty}(2^{-k} \cdot k) &= 2^h \cdot \frac{1/2}{(1-1/2)^2} \\ &= 2 \cdot 2^h \le 2n\end{aligned}$$

where we have used that the number of nodes in a complete binary heap is $2^{h+1} - 1$ hence $n \ge 2^h$ as the levels $1, \ldots, h-1$ are full and there is at least one node at the level h.

5 Graph transitive closure

In the following chapter we consider the problem of finding the transitive closure (the list of accesible nodes) for Directed Acyclic Graph (DAG).

Algorithm 1: Topological sort for DAG

```
Data: G = (V, E) – DAG, |V| = n < ∞, v – starting node, visited ← {}
Result: ℓ – the order of topological sort
add v to visited;
for u ∈ V : (v, u) ∈ E do
    if u ∉ visited then
        topological_sort(G, u, visited);
add v to ℓ;
return ℓ_n, ..., ℓ_1
```

Intuition for Algorithm 1. By definition of the topological sort we must have that for each edge $v \to u \in E$ the vertex v must come before u. This is the immediate result of the algorithm construction since for each vertex we first run the topological sort from all of its children meaning that we add them before the vertex v. Since we maintain the list of the visited nodes we have that each node is visited at most ones, which also implies that the algorithm converges as the number of vertices is finite □

Lemma 5.1. *Given a DAG $G = (E, V)$ we have that:*

$$R(v) = \bigcup_{u \in V:(v,u) \in E} R(u) \cup \{v\}$$

where $R(v)$ denotes the set of nodes accesible from v

Intuition for Lemma 5.1.

"$\supset$" That part is trivial since we can go to any node u with the direct edge from v or stay in v and then to any node within $R(u)$

"$\subset$" Suppose the node u is reachable from v. Assume that $v \neq u$ (the case $v = u$ follows trivially). Let $v = w_0, w_1, ..., w_k = u$ denote the path from v to u, i.e. $(w_i, w_{i+1}) \in E \; \forall i \in \{0, ..., k-1\}$. Truncating the path (as we assume $v \neq u$ meaning that $k \geq 1$) to $w_1, ..., w_k$ we get that $u = w_k \in R(w_1)$, which implies:

$$u \in R(w_1) \subset \bigcup_{w \in V:(v,w) \in E} R(w)$$

□

Algorithm 2: Transitive closure

Data: $G = (V, E) = (\{v_1, ..., v_n\}, E)$ – DAG, $v_k \in R(v)\ \forall k \in \{1, ..., n\}$,
$R(v_k) \leftarrow \{v_k\}\ \forall k \in \{1, ..., n\}$

Result: $R(v_1), ..., R(v_n)$

```
ℓ ← topological_sort(G, u);
reverse ℓ;
for u ∈ ℓ do
    for w ∈ V : (u, w) ∈ E do
        R(u) ← R(u) ∪ R(w);
        /* Note that w was processed before since in the reversed topological
           order u comes after w for any edge (u, w)                        */
return R(v_1), ..., R(v_n)
```

Remark 5.1. The Algorithm 2 merely implements the Lemma 5.1